



Leveraging AI-enabled knowledge capabilities for new product performance: Evidence from manufacturing SMEs in emerging economies

Van Hoang Dinh ^{*}

Foreign Trade University (FTU), Hanoi, Viet Nam

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ABSTRACT

Emerging market manufacturing small and medium-sized enterprises (SMEs) struggle to translate artificial intelligence (AI) investments into innovation outcomes. This study investigates how AI-enabled knowledge capabilities drive new product performance by employing a sequential mixed-methods design with 261 manufacturing SMEs. Our findings reveal that AI-enabled knowledge capabilities operate through dual pathways: directly enhancing performance and indirectly through digital product innovation radicalness. Furthermore, this study identifies digital sustainability leadership as a critical boundary condition acting as both mediator and moderator, fundamentally reconfiguring how AI generates value. This study makes significant contributions to dynamic capabilities theory and the knowledge-based view. By positioning AI-enabled knowledge capabilities as critical dynamic capabilities, this research clarifies the transformation pathways through which AI generates competitive advantages and offer actionable insights for managers navigating resource-constrained environments.

1. Introduction

The contemporary manufacturing landscape confronts an existential imperative: traditional knowledge management no longer suffices to drive innovation in emerging markets. While the global knowledge economy generates over \$9 trillion annually (U.S. National Science Foundation, 2020), emerging market manufacturing SMEs struggle disproportionately to convert knowledge assets into new product performance (Kumar et al., 2020). A troubling implementation-performance gap persists: although AI fundamentally transforms knowledge processes, potentially unlocking \$2.3 trillion for emerging markets (World Economic Forum, 2023), most SMEs fail to realize the 40% higher success rates achieved by AI leaders (Rezaei, 2025). This disconnect reveals a critical theoretical void regarding how AI-enabled knowledge capabilities function as dynamic capabilities in resource-constrained contexts (Braganza et al., 2021; Priyono et al., 2020).

Prior research has examined knowledge management extensive, yet findings regarding the knowledge-performance relationship remain inconsistent. While some studies identify direct effects (Joshi & Sharma, 2004; MacCurtain et al., 2010) or mediating mechanisms like knowledge integration (De Luca & Atuahene-Gima, 2007; Su et al., 2013), others point to conflicting moderators such as technological turbulence

(Jin et al., 2019; Tsai & Hsu, 2014) or inverted U-shaped engagement effects (Yepeng et al., 2022). Furthermore, research on AI's role remains polarized. While AI expands knowledge depth (Khalid et al., 2025) and facilitates innovation stages (Cooper, 2024), it also introduces “dark side” risks, including knowledge hiding (Arias-Pérez & Huynh, 2023), leakage (Ritala et al., 2015), and algorithmic gatekeeping (Keegan et al., 2023; Papagiannidis et al., 2023). Existing frameworks, largely developed in pre-AI contexts (Grant, 1996; Teece, 2007), inadequately explain how AI transforms sensing and seizing mechanisms in volatile environments (Mikalef & Gupta, 2021; Raisch & Krakowski, 2021).

To address these gaps, this study grounds itself in dynamic capabilities theory and the knowledge-based view, integrating practice-theoretical perspectives (Whittington, 2006). This study proposes that AI-enabled knowledge capabilities do not automatically generate advantage but require specific transformation pathways (Schilke et al., 2018; Teece, 2018). Specifically, this study identifies digital product innovation radicalness - breakthroughs utilizing digital technologies (Nambisan et al., 2017; Pesch et al., 2021) - as a critical mediator. This responds to calls to explicate value creation mechanisms in AI-enabled systems (Kohtamäki et al., 2025; Sjödin et al., 2021). Furthermore, this research examines digital sustainability leadership as a boundary condition. As firms face dual pressures for digitalization and

^{*} Corresponding author at: 91 Chua Lang Street, Lang Ward, Hanoi 100-000, Viet Nam.

E-mail address: dinhvanhoang@ftu.edu.vn.

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